आकार

AKAR

A system of surfaces and joineries, which makes easy assembly of doubly curved surfaces a reality



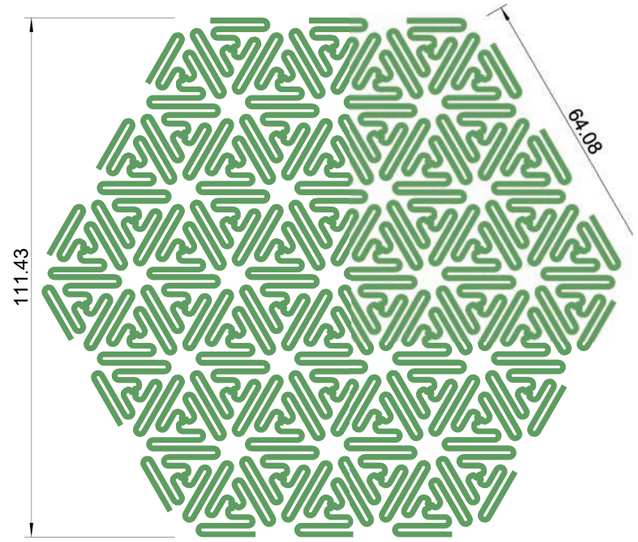
Part 1

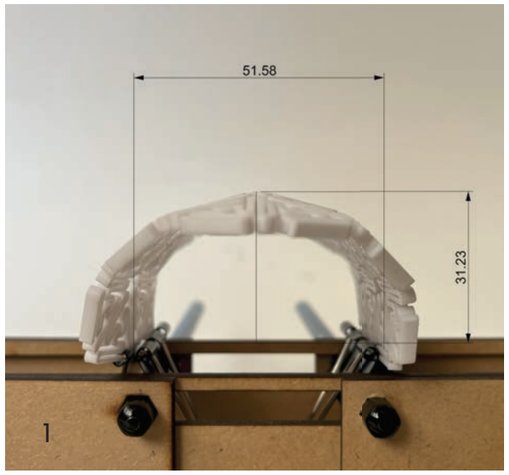
Creating a System

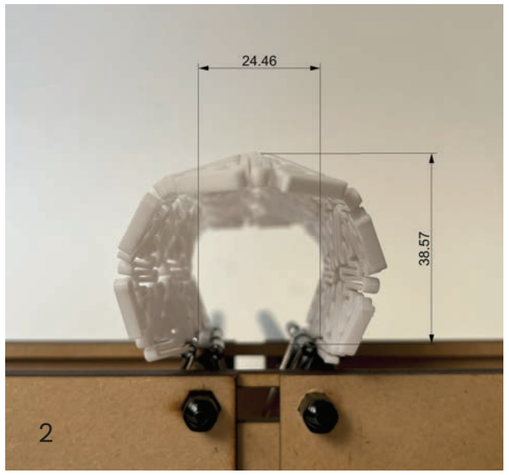
- Doing a series of experiments using 3d printing
- Analyzing the results to understand physical properties
- Creating a system based on the understanding

Experiment

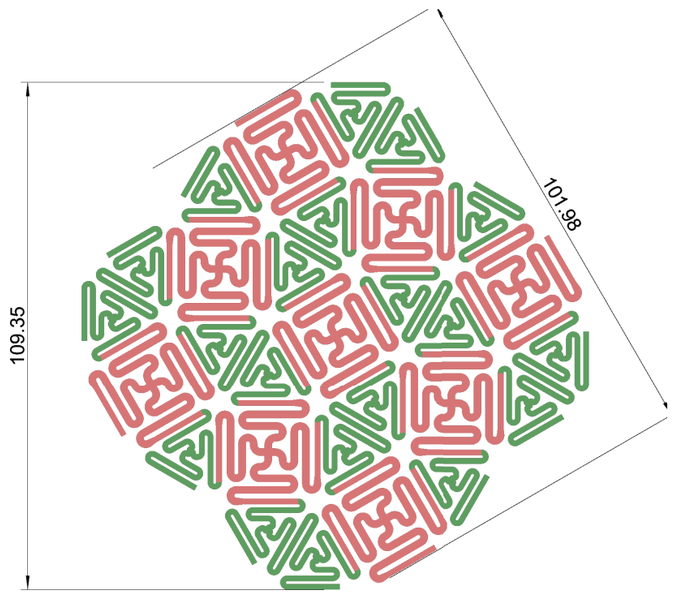
Doubly Curved Surface

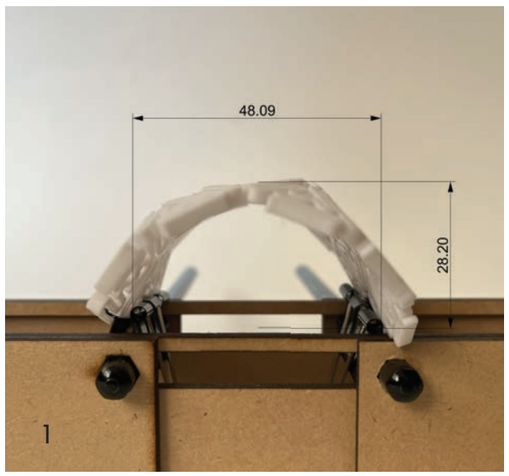


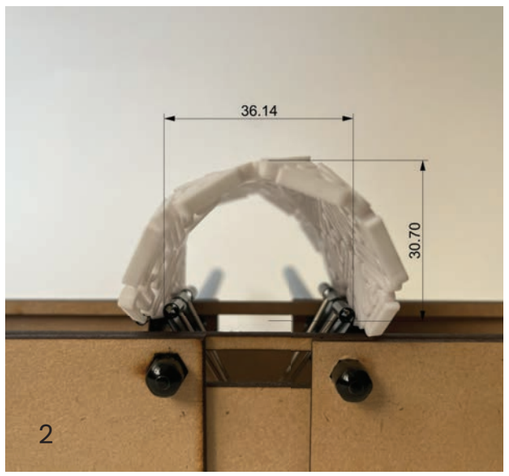




	1	2
Height	31.23	38.57
Base	51.58	24.46
Ratio	0.61	1.57



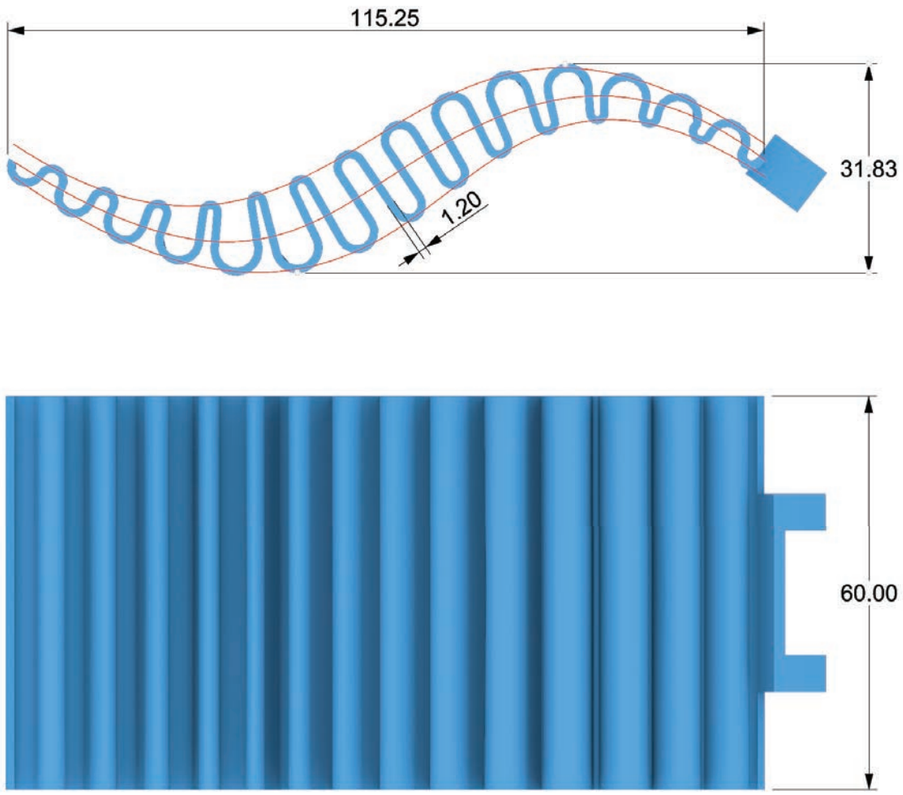


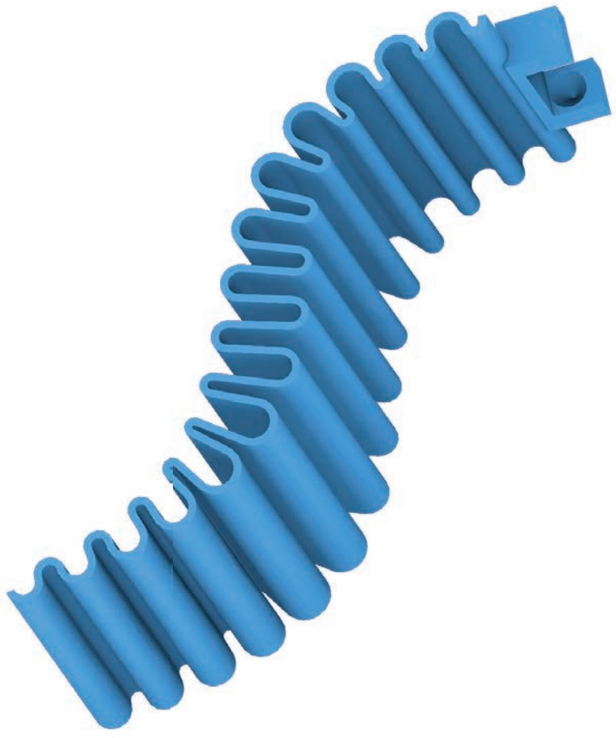


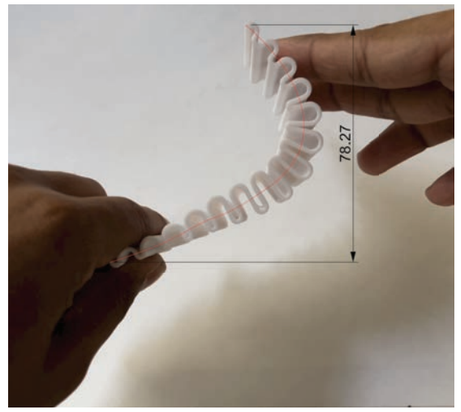
	1	2
Height	28.20	30.70
Base	48.09	36.14
Ratio	0.58	0.85

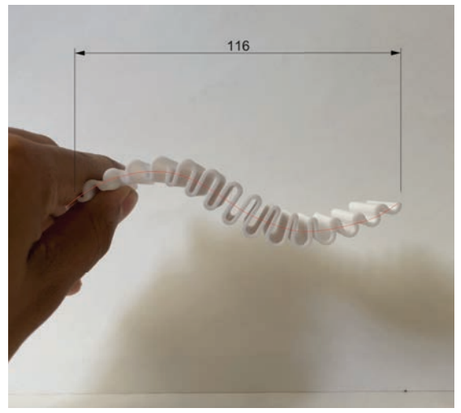
Experiment

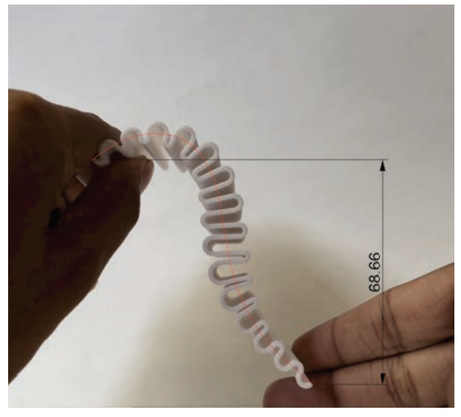
Irregular Extrusions

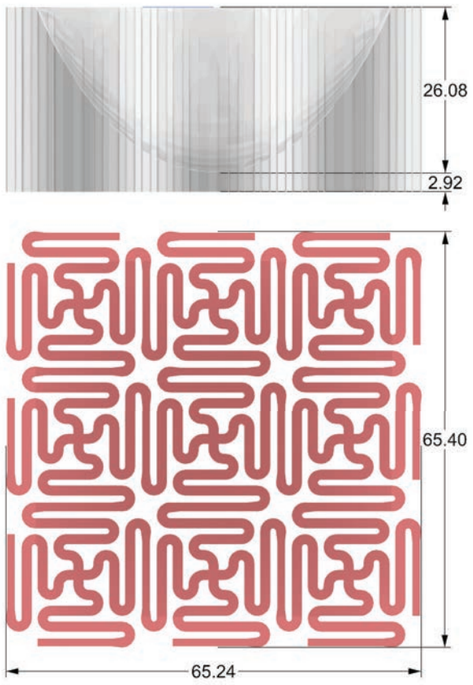


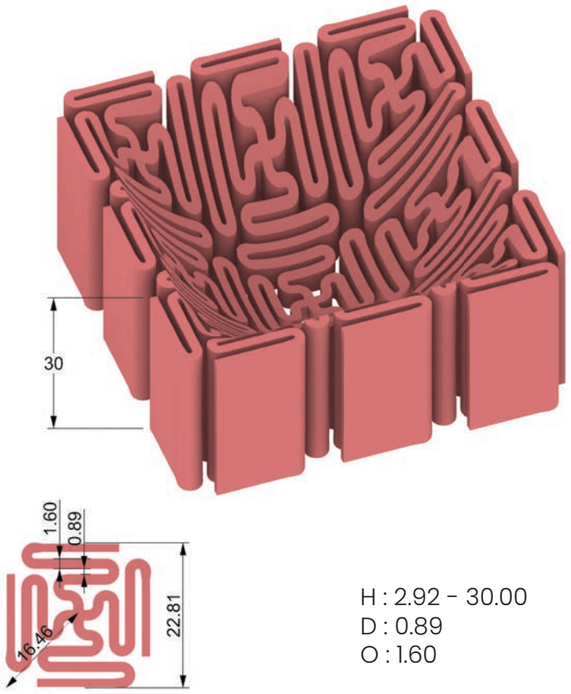






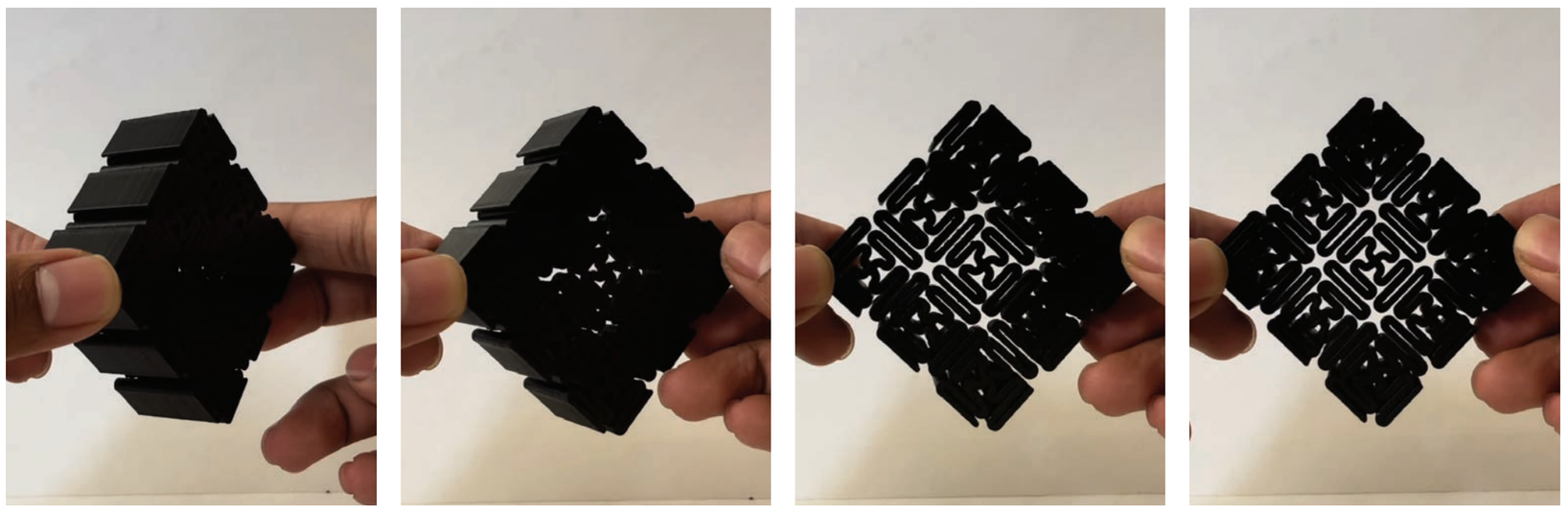






H : 2.92 - 30.00
D : 0.89
O : 1.60

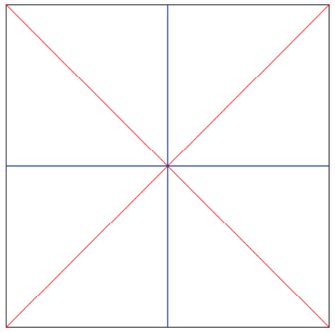
Test : Visibility



Angle: 25 45 75 90

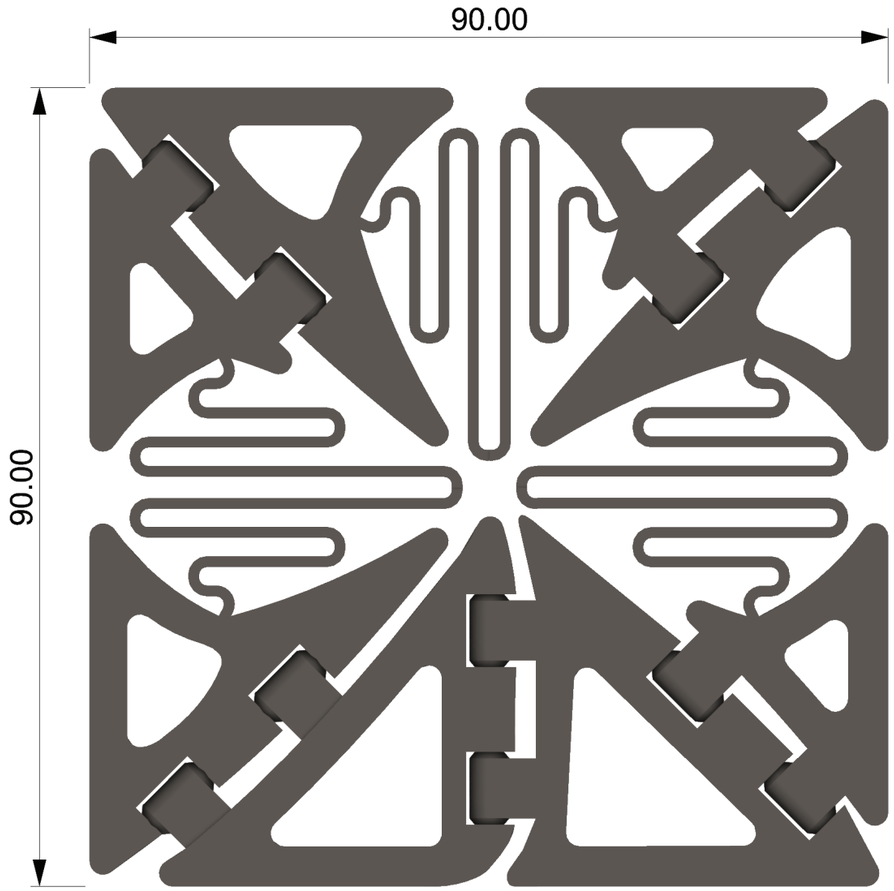
Experiment

Print in Place Hinges



Crease Pattern : Water-bomb Base

Red - Mountain Fold
Blue - Valley Fold



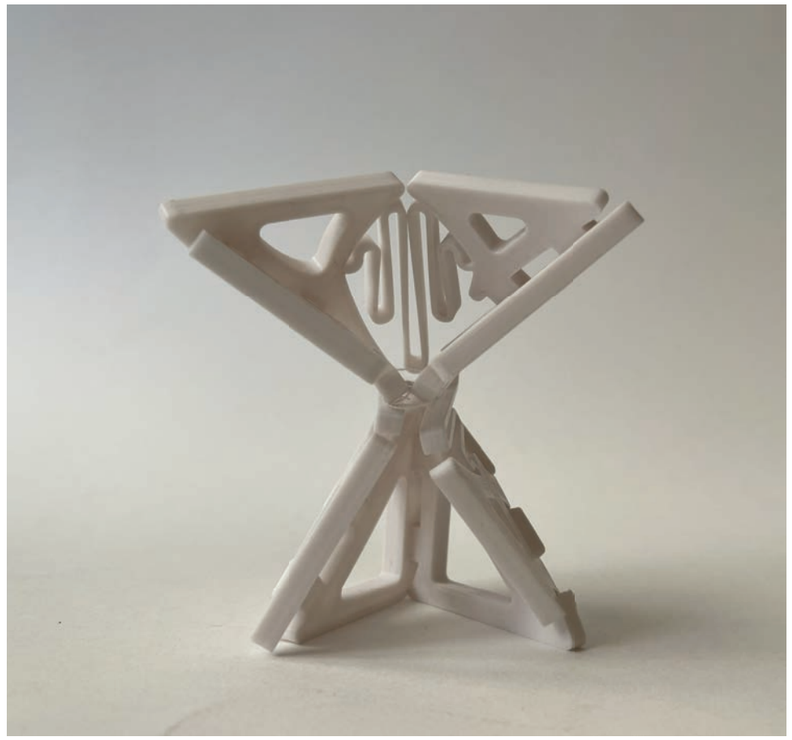
Spring

H : 4.800 mm
D : 2.427 mm
O : 1.200 mm

Hinge

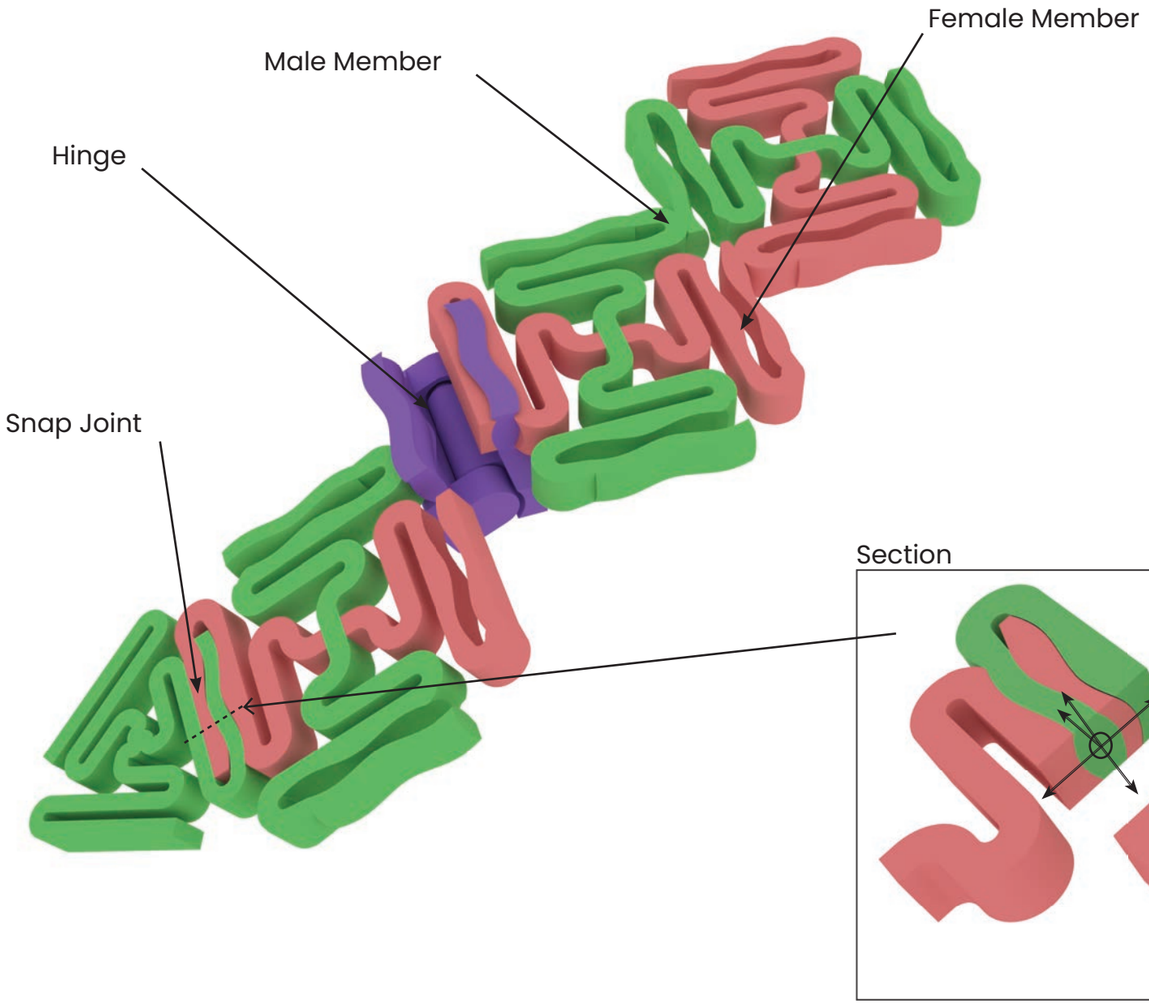
T : 0.480 mm
H : 1.907 mm
W : 6.750 mm
L : 7.500 mm



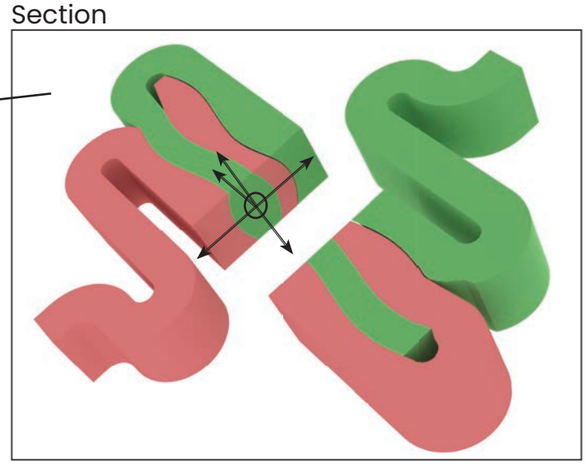


Experiment

Modularity 1



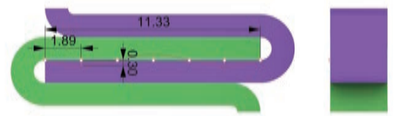
Section



Motion Restricted in all the directions



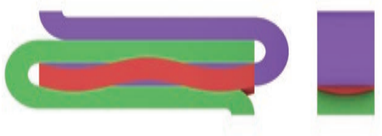
Inspiration



Step 1



Step 2



Step 3



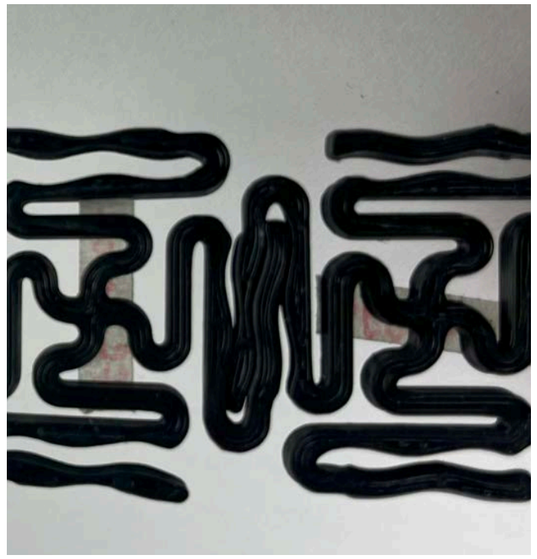
Step 4



Rigid joinery



Iteration with different tolerances



Detail of joinery



Iterations of Removable Hinge



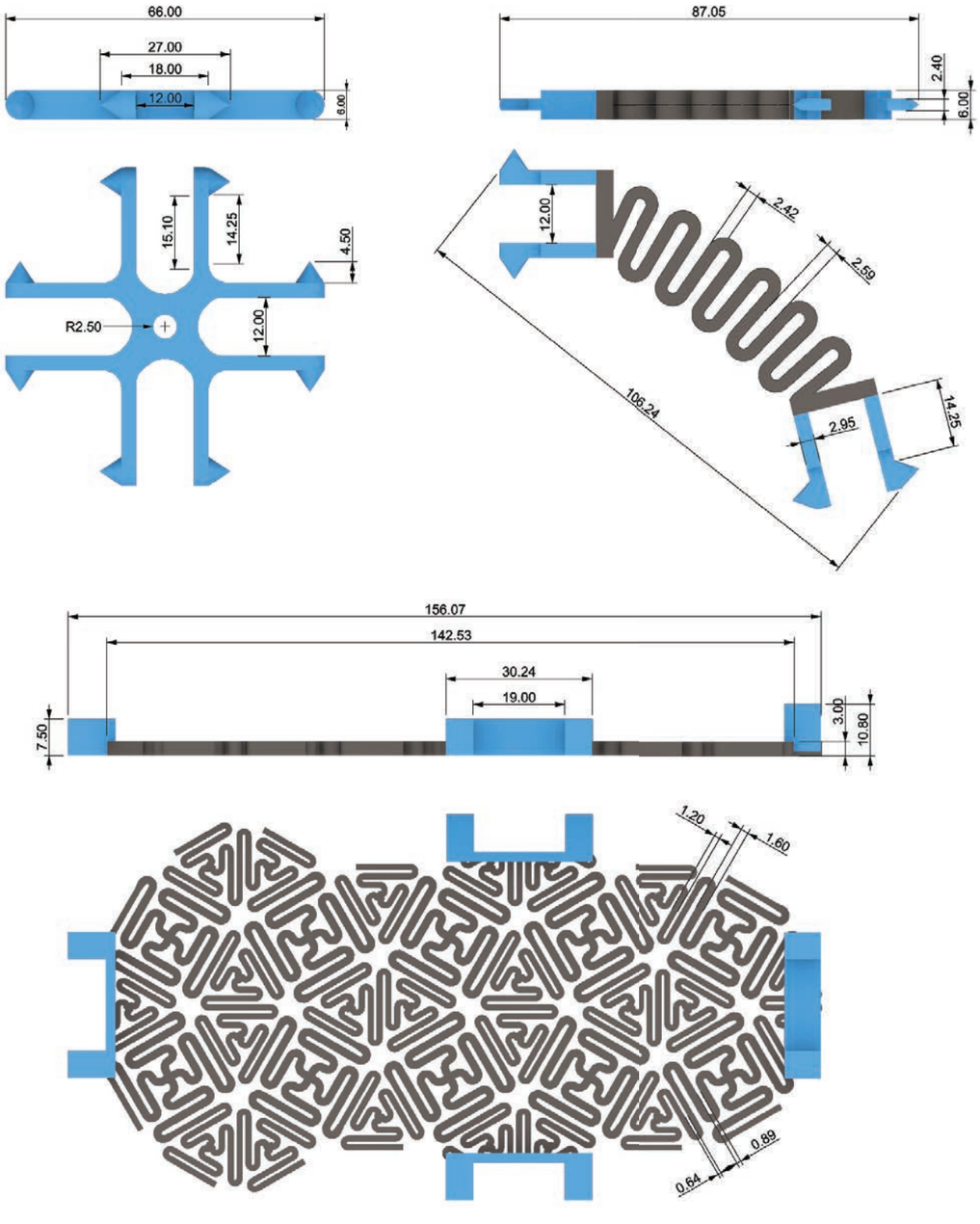
Perspective View



Assembly using hinges

Experiment

Modularity 2



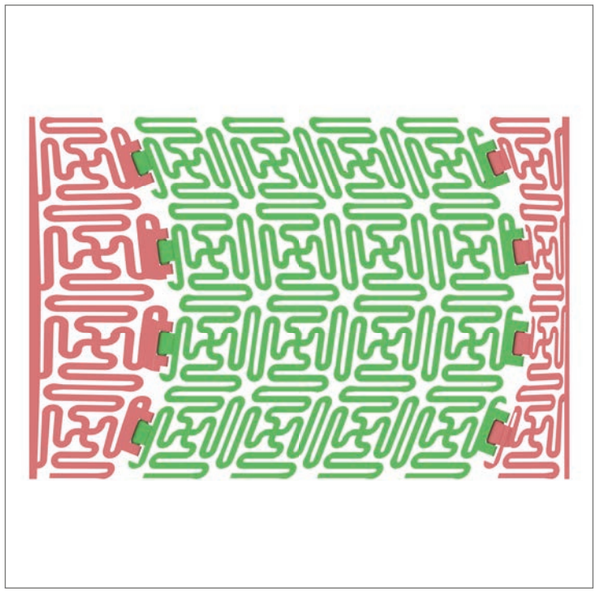




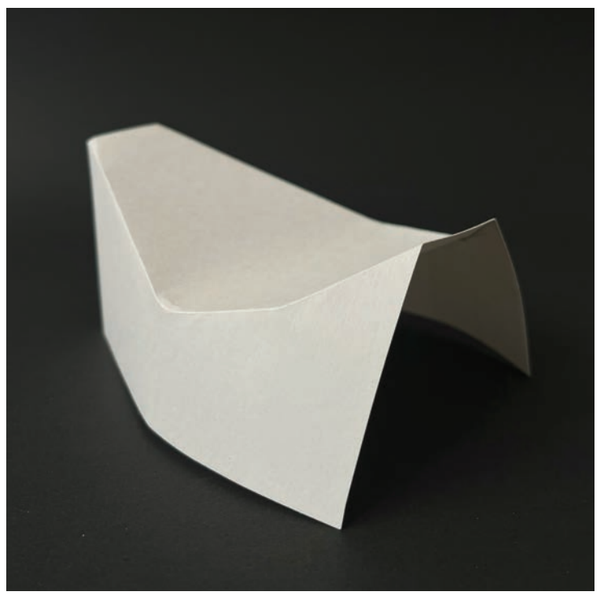
Experiment

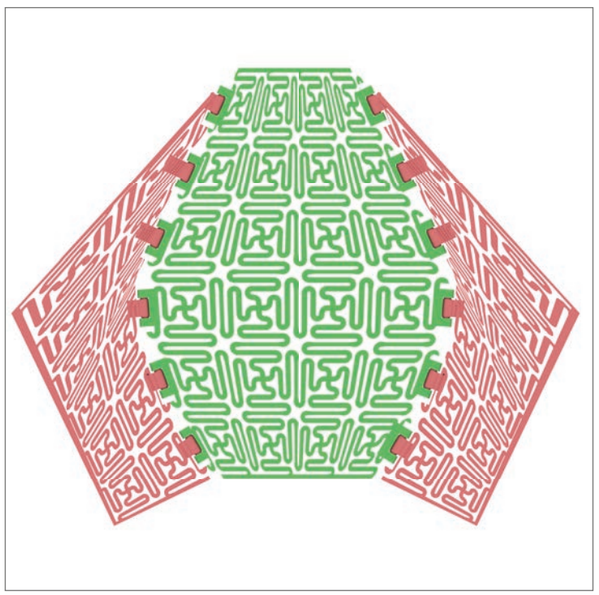
Surface Tension

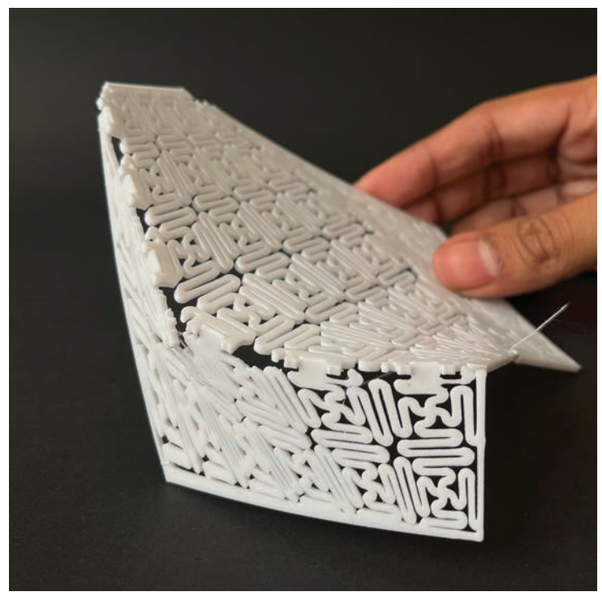












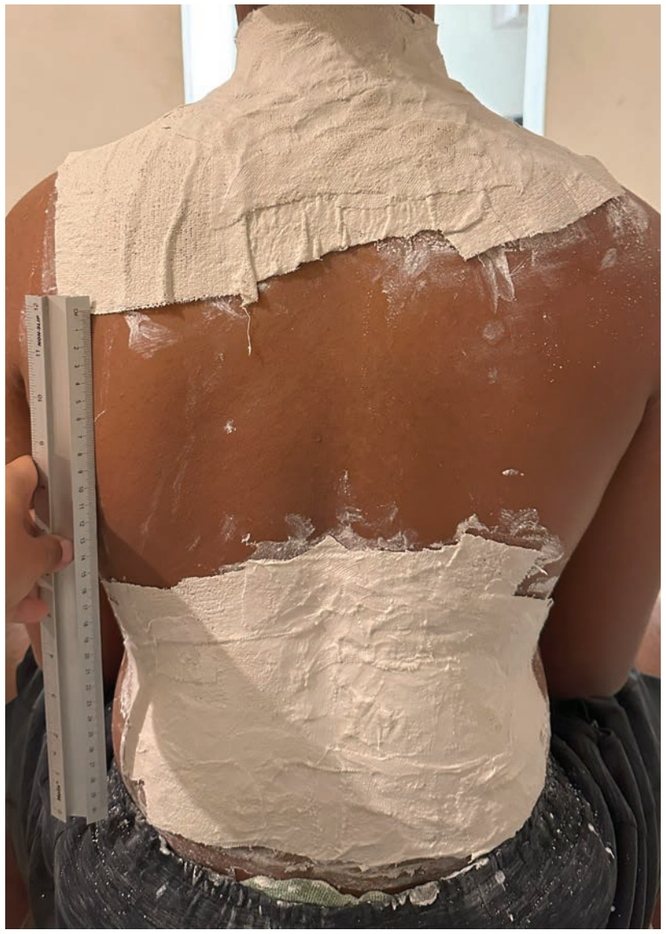
Part 2

Application of the आकार System

- Choosing a domain
- Anthropometric study
- Form development
- Optimization of the system
- Prototype

Anthropometric Study

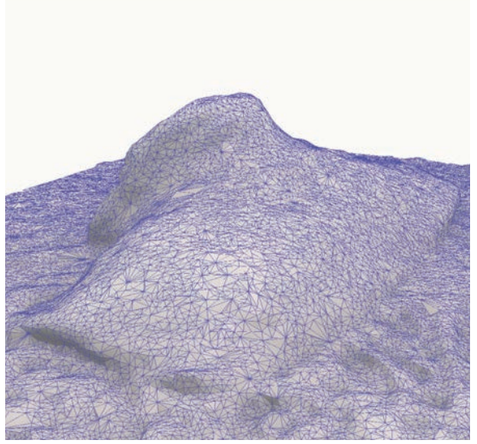
Casts Created Using POP Plaster



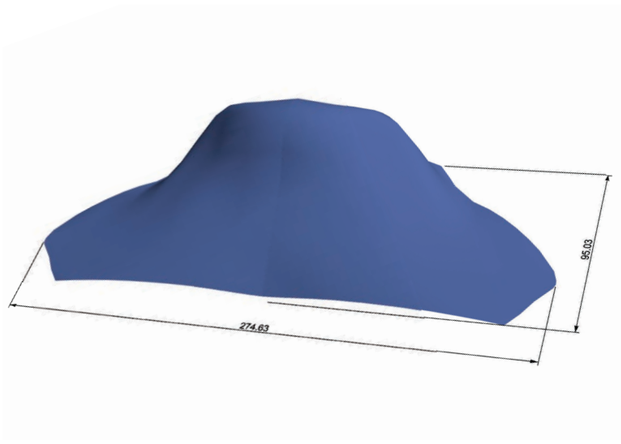
realistic scan



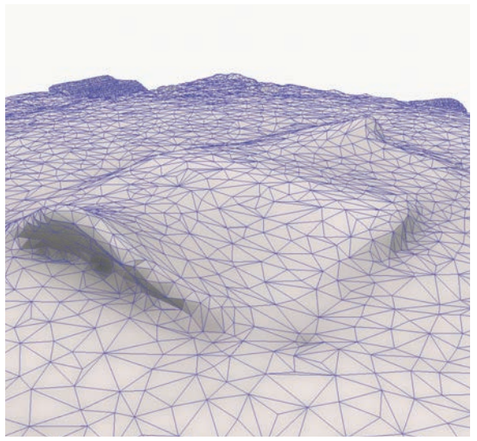
mesh

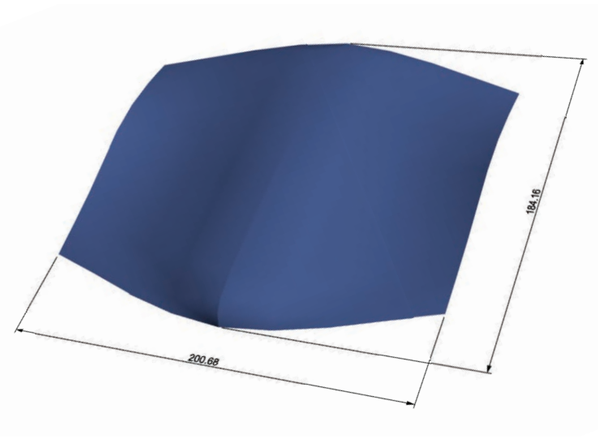


Final Surface

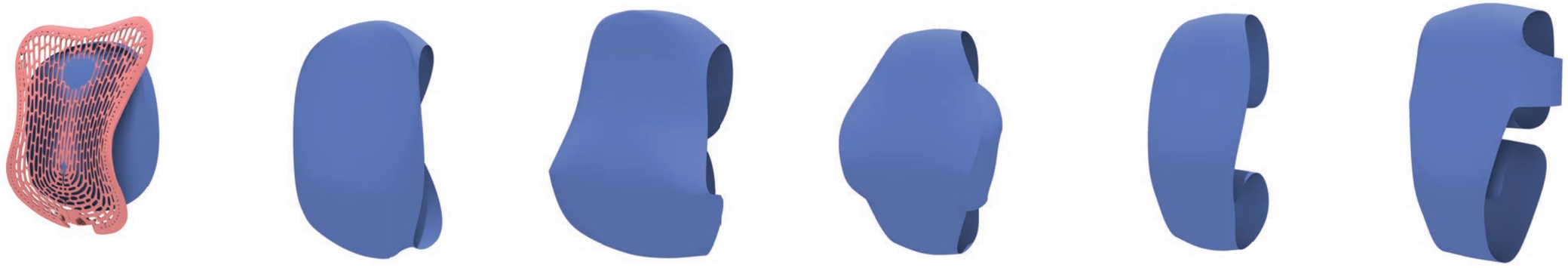


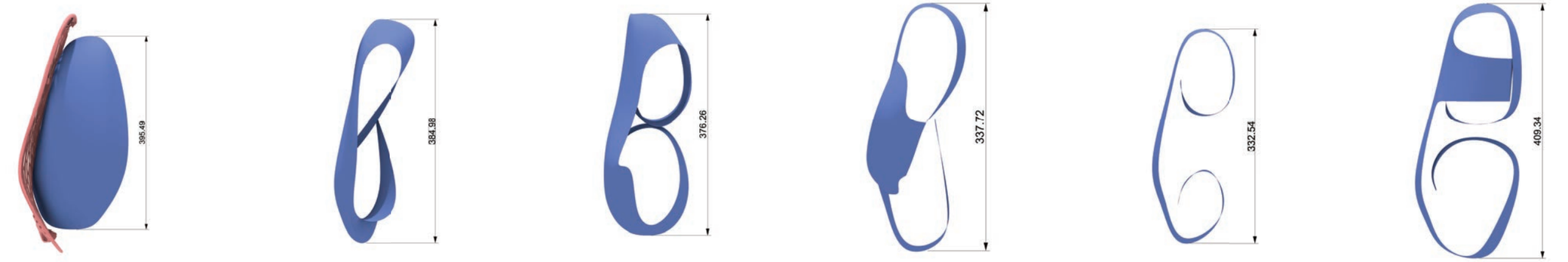


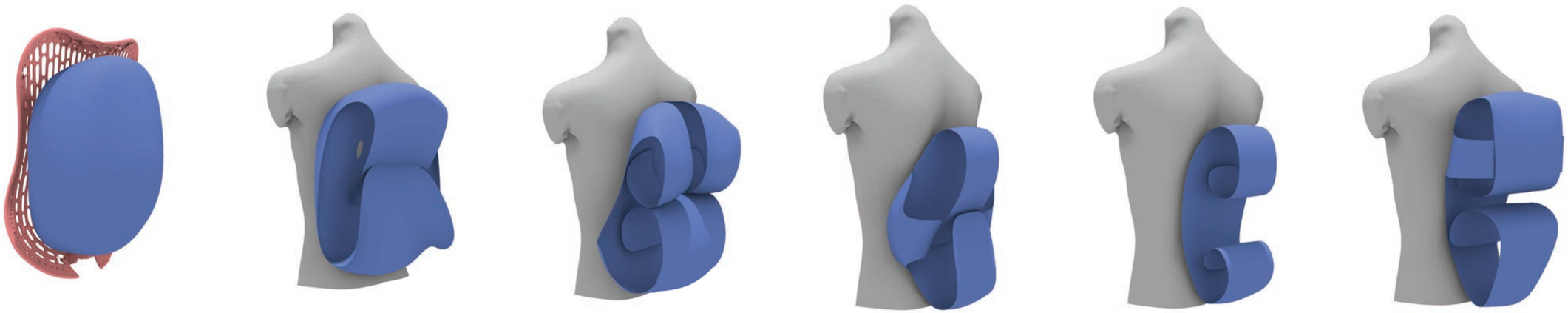


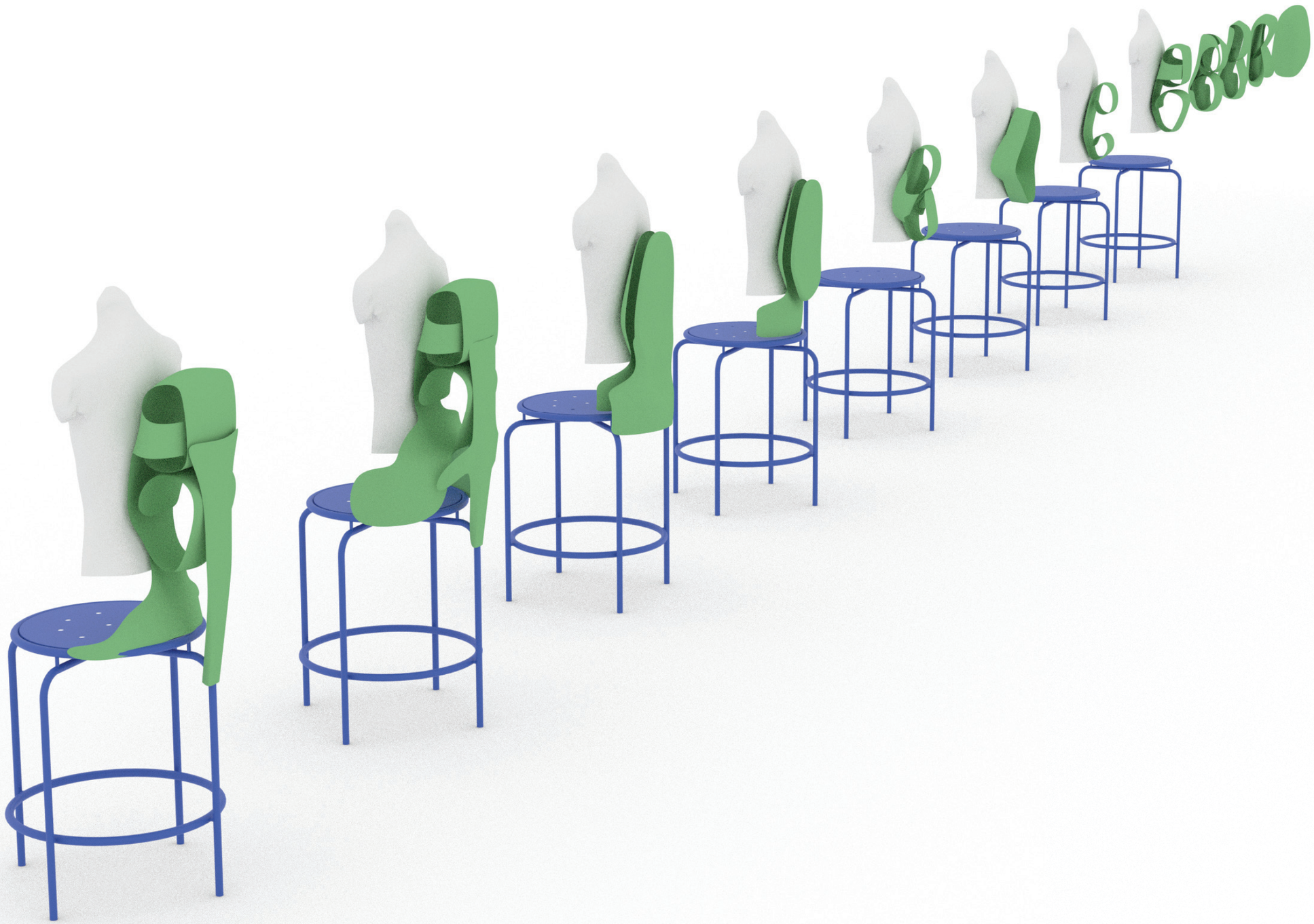


Form Development









Cardboard Model



Sitting without reclining

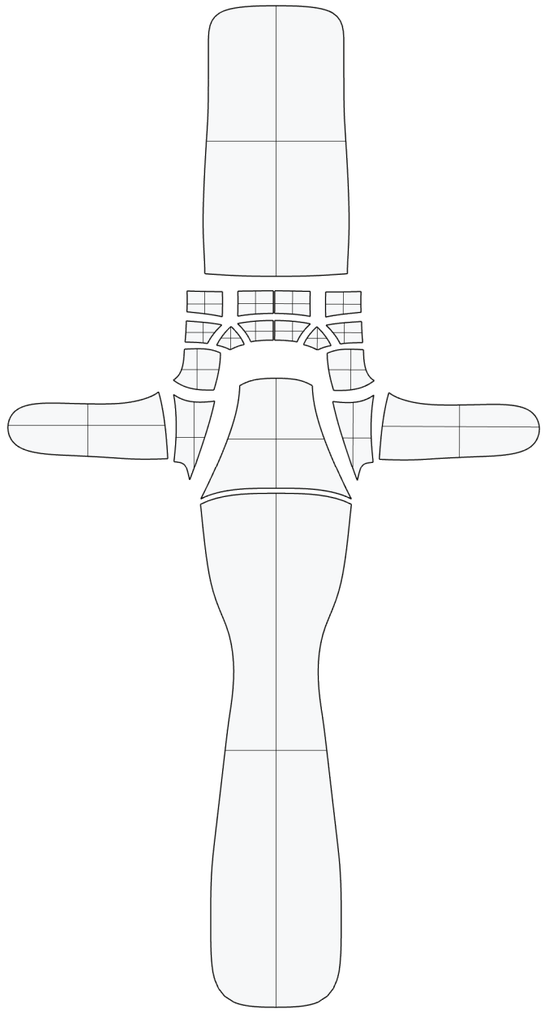


Sitting upright

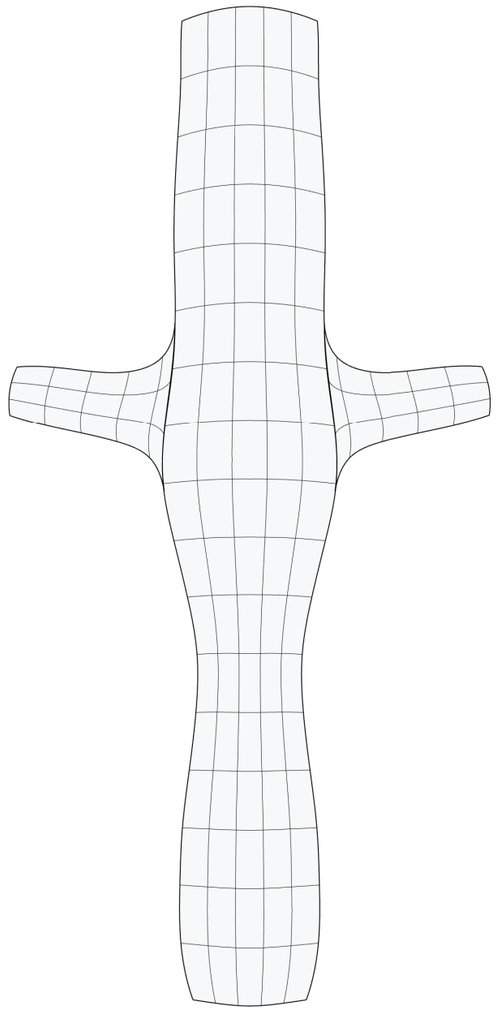


Fully Reclined

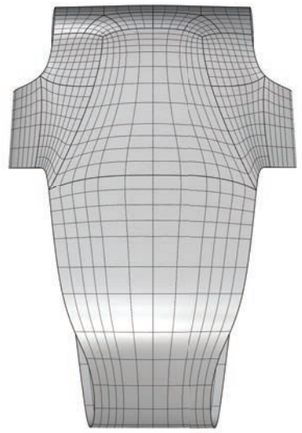
Surface Optimization



Unrolled fragmented surface



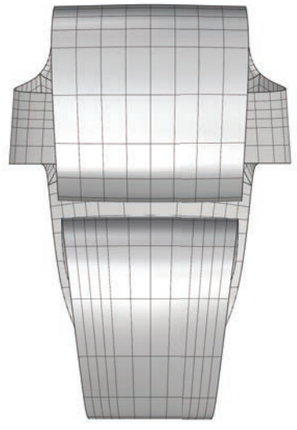
Final Surface



Front

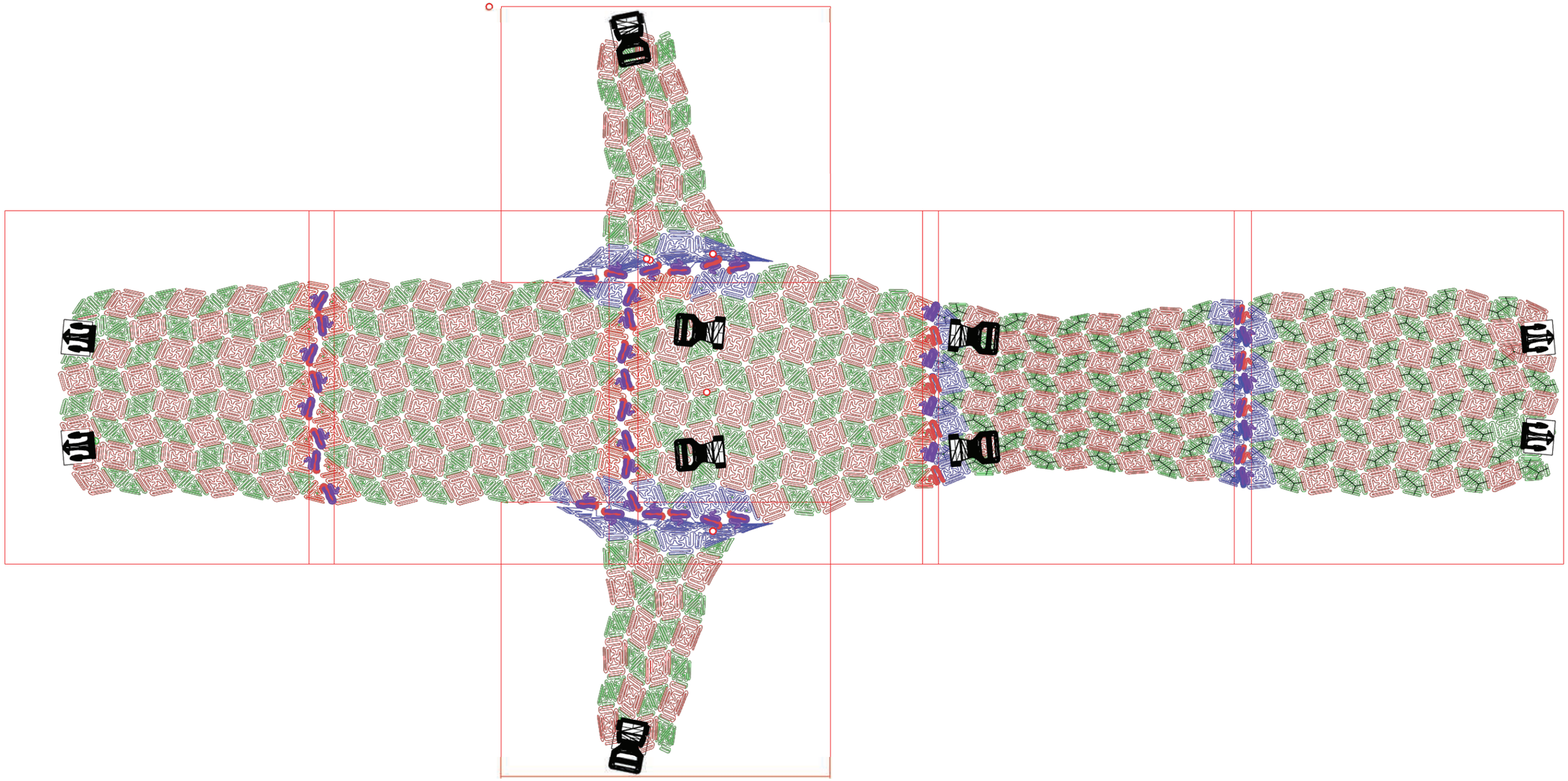


Right



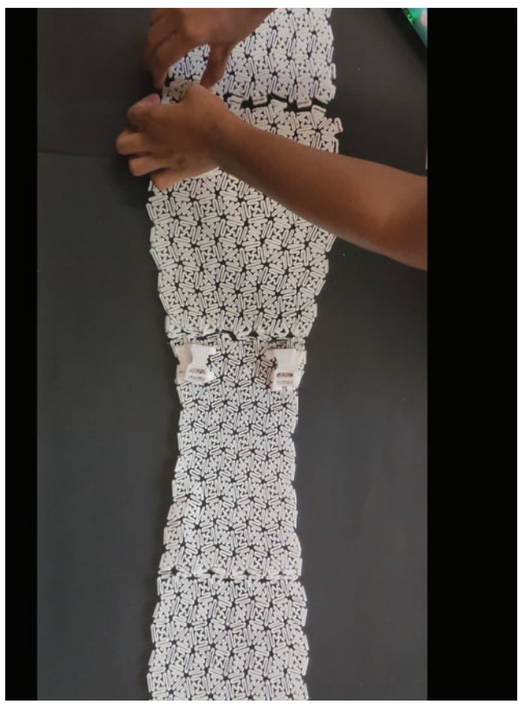
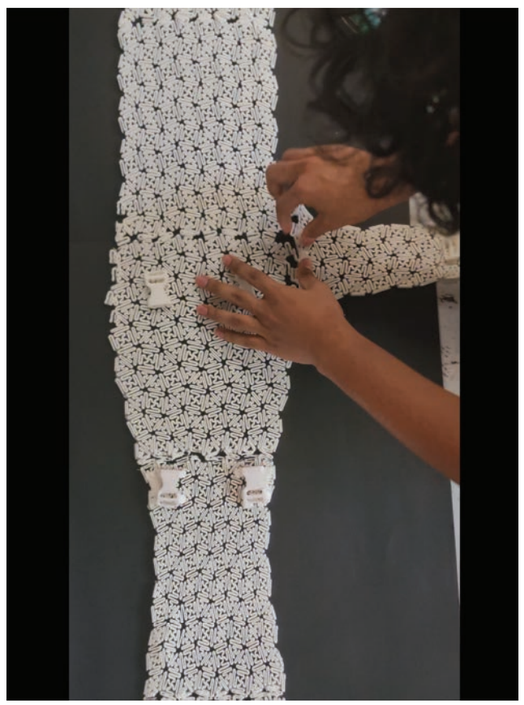
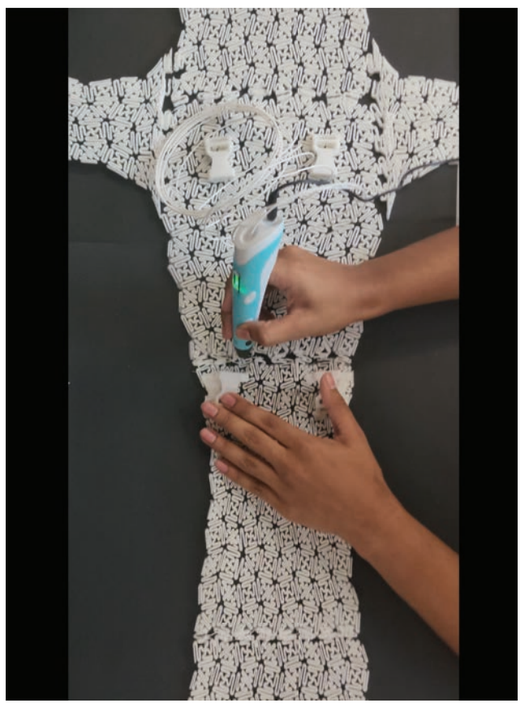
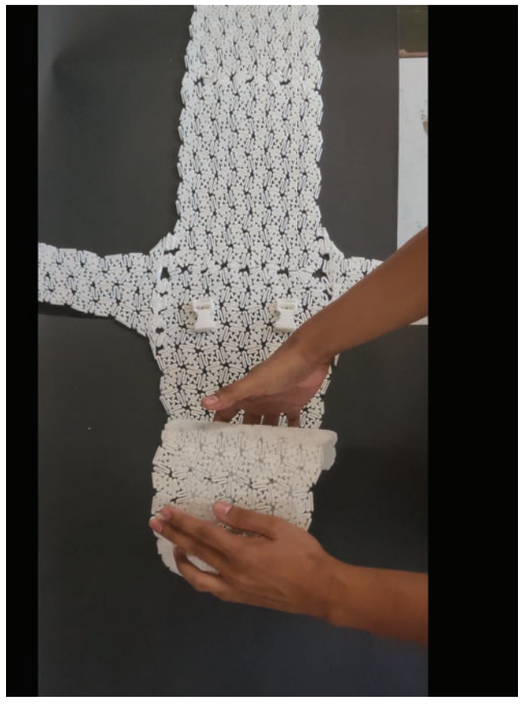
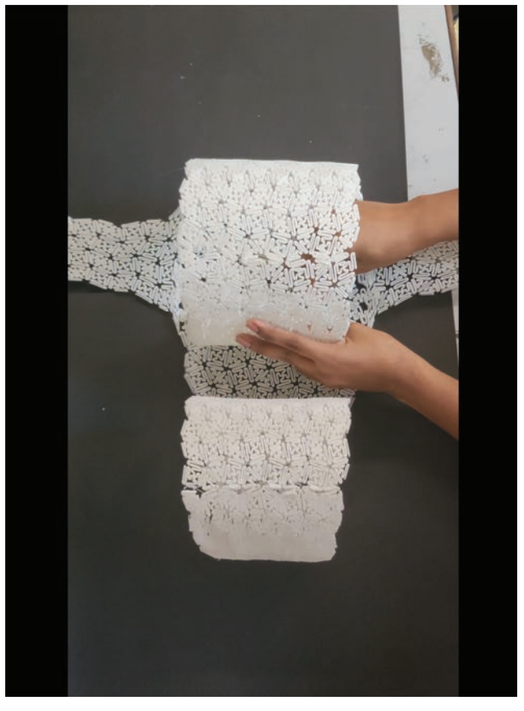
Back

System Optimization



The full form resolved in form of the “Akar” system

Assembly



Prototype



Upright



Reclined



